

The Pandemic Trilemma

When thinking back to 2020, the word COVID comes instantly to our mind. What a time this was, defined by chaos, fear, and some really weird situations. Governments had to step in to control this unique situation without much prior knowledge or experience, while trying to balance a threefold problem: protecting human lives, preserving economic activity and the livelihoods that depend on it, and maintaining fiscal sustainability to safeguard the state's capacity to act. As the virus was lethal and no solution to it was in sight, the protection of human lives was only feasible through contact reduction in several different ways. Those restrictions, however, simultaneously threatened the second sphere of the problem: economic activity. Governments therefore implemented a wide range of highly diverse fiscal support measures to cushion the economic damage, while at the same time putting fiscal sustainability under pressure. The result: substantial differences in outcomes across all three dimensions. So, what drove this heterogeneity? Which fiscal instruments worked best, and which policy composition was most effective? Answering this question provides both a retrospective assessment of pandemic policy and a benchmark for future crises in which governments must allocate fiscal support under binding health, economic, and budgetary constraints.

My research question therefore is: *What is the optimal composition and timing of fiscal support during a pandemic, given the epidemiological state, and what are the costs—in foregone output stabilization and excess debt—of deviating from this benchmark?* To study this question from an economic perspective, five elements have to be understood. i) The pandemic environment itself: how can the unique situation be characterized ii) The policy response: which containment and fiscal policy measures did governments implement, and how can these measures be classified into economically distinct instruments iii) The transition effects: how did each policy instrument affect health, output, and debt, and by what magnitude iv) The interaction between these dimensions: how did containment, fiscal support, and epidemiological dynamics jointly shape the trade-off between health stabilization, output stabilization, and fiscal sustainability v) The policy benchmark: given these estimated transitions, which strategy would have performed best in managing the threefold problem.

Therefore, I proceed in two steps: i) I formalize the pandemic situation in a reduced-form theoretical framework from the perspective of a social planner. Building on the macroeconomic literature, I bring the health, output, and debt dimensions together and define the channels through which different policy measures affect them. This shows that the pandemic policy problem is, in theory, a trilemma. I then bring the resulting transition system to the data for OECD countries, combining publicly available macroeconomic data (real GDP, real debt), containment-policy indicators (Oxford Stringency Tracker), epidemiological variables (excess mortality, infection rate, cases), and a novel database that I collected on individual fiscal measures implemented during the pandemic and aggregated to the quarterly level for the period 2020–2022. The fiscal-measure database contains 2,301 individual policy measures, which I classify into the fiscal instruments defined in the model: capacity preservation which stabilizes firms, employment relationships, and existing economic structures, and demand injection, which provides an impulse to consumption but is more questionable in a supply-constrained economy. These instruments differ in timing, target group, budgetary treatment, and transmission channel. This data collection was needed because existing fiscal trackers do not provide the granularity required to assess these economic channels. ii) Empirically, I estimate the transition equations on a quarterly OECD panel for 2020–2022. The estimated parameters show how containment and different fiscal instruments map into the output gap, excess mortality, and the pandemic debt gap, and whether the trilemma is present in the data. I then use them to calibrate a social-planner optimization problem in which he minimizes losses across all three dimensions. Solving this problem provides optimal policy paths for the OECD average and for country- or group-specific averages, allowing me to assess the overall pandemic strategy.

This research design is not intended to estimate standard causal treatment effects of individual policy measures or to provide a fully microfounded model of agent behavior. Instead, the reduced-form approach deliberately abstracts from individual optimization and focuses on the aggregate transition channels between containment, fiscal support, health outcomes, output, and debt. In a dynamic

panel setting, governments respond to current and expected pandemic conditions, so strict causal identification is difficult. I therefore interpret the estimates as empirical transition parameters rather than as standalone causal effects. Nevertheless, the estimates show whether the model-implied channels are present in the data, whether they have the expected signs, and what magnitudes they imply. This is appropriate for the purpose of the paper: the objective is not to isolate one policy shock, but to discipline a dynamic system for counterfactual policy evaluation. While the broad COVID literature has written extensively about trade-offs between health and economic activity, fiscal measures, and the micro-founded macroeconomic effects of specific interventions, this integrated approach remains to my best knowledge missing. Therefore I address this gap by linking these three dimensions in one estimated transition system, classifying fiscal measures according to their economic channels, and using the resulting model to assess which policy composition would have best managed the pandemic trilemma.

As I have already made substantial progress with this project, I provide here a short summary of what I have already done, what still remains to be completed, and some first results, in order to maximize the quality of feedback I can receive during this course.

So far, I have developed the theoretical trilemma framework, defined the relevant health, output, and debt states, and derived the corresponding transition system. I have also constructed a novel fiscal-measure database covering 2,301 individual pandemic measures across all 38 OECD countries and classified them by economic transmission channel. In addition, I have assembled the quarterly OECD panel, defined the main variables, and estimated the output and debt transition equations, which show that the trilemma is empirically present, with the key transition parameters statistically significant and showing the expected signs and magnitudes. I have also parameterized the health-side block and validated the calibrated transition system against observed pandemic dynamics. First results indicate that a policy path exists that dominates the OECD-average strategy, with a clear timing, composition, and magnitude of fiscal instruments.

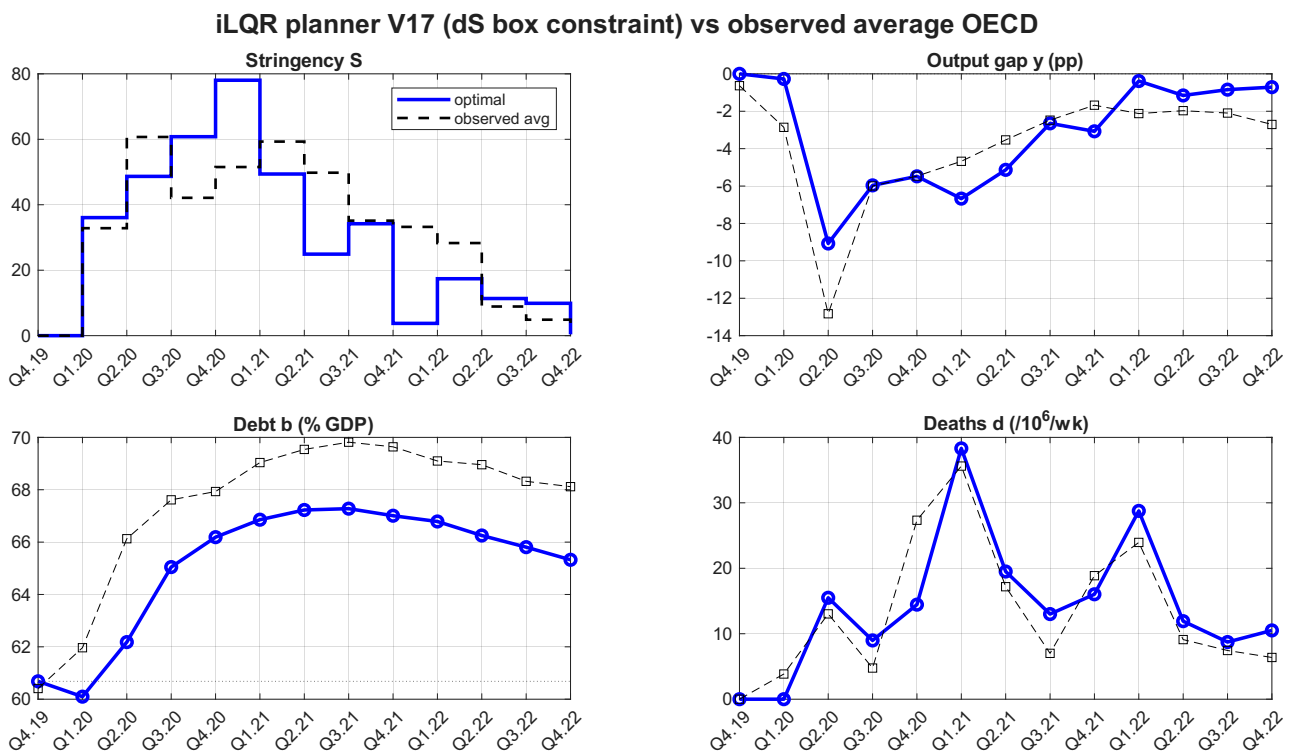


Figure 1: First Results Optimization